TimeJEPA: What 1.1M Parameters and Joint Embeddings Can Do

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Abstract

TimeJEPA is a 1.14M-parameter univariate forecaster pretrained with a joint embedding predictive architecture (JEPA), evaluated zero-shot on all 97 GIFT-Eval configurations with one checkpoint, one configuration and no per-dataset adaptation, and trained on three consumer GPUs. It reaches 0.863 / 0.596 (MASE/CRPS ratios vs seasonal naive with sign-flip TTA; 0.895 / 0.624 without), between Moirai-large (311M) and Moirai-base (91M) on CRPS. Three claims, each carried by paired, pre-registered experiments. (1) The accuracy comes from corpus composition and inference protocol: a one-variable ablation finds latent extrapolation indistinguishable from reconstruction, and a per-step analysis refutes its apparent long-horizon advantage. (2) The frozen JEPA latent is nonetheless a competent judge of candidate futures: as a zero-training reranker it turns TTM-R3 into a probabilistic forecaster that beats its own weighted quantile loss on 6/6 held-out datasets and beats its point CRPS by 7% over all 97 GIFT configurations on paired windows. (3) Four latent dimensions of predicted residual statistics carry the calibration of the forecast distribution: disabling their gate costs 18.7 CRPS points at a bit-identical median, and the counterfactual trained without them matches the champion's CRPS, so what the latent buys is nominal coverage at equal accuracy. The corpus makes the forecast; the latent makes the judge. Clean negative results are reported as results.

1 Introduction

We tested latent extrapolation as a pretraining objective for forecasting; at equal corpus and budget it is indistinguishable from reconstruction (Section 6). Given the current interest in JEPA for time series [3, 21], we state this plainly, and report what the project delivers instead (Figure 1):

- **A competitive small foundation model, from data alone.** One 1.14M checkpoint, one configuration, all 97 GIFT-Eval configurations zero-shot: 0.863 / 0.596 with flip TTA (0.895 / 0.624 plain). Every accuracy step traces to corpus composition or inference protocol; none to the objective (Section 5).
- **The frozen latent is a judge.** Its implicit energy ranks true futures far above chance and, as a zero-training reranker, turns TTM-R3 [12] into a probabilistic forecaster that beats its own WQL on 6/6 held-out datasets and its point CRPS by 7% across GIFT (Section 7).
- **Four latent dimensions carry the calibration.** A residual statistics head, gated into the quantile decoder, is load-bearing for the predictive distribution: gate off costs +18.7 CRPS points with MASE bit-identical, and what it buys the full model is nominal coverage at equal accuracy (Section 7).

Three properties frame every number: one checkpoint and one configuration for all 97 configurations (the TTM family, among others, publishes per-regime variants; we do not); nothing tuned or selected on GIFT-Eval data; three RTX 3090s end to end, with predictions registered in a dated log before each run and negatives reported as results (Section 8).

2 Related work

Zero-shot forecasting spans several architectural paradigms, none dominant: decoder-only (TimesFM [11], Toto [10]), masked encoders (Moirai [29], whose LOTSA corpus we pretrain on), tokenized seq-to-seq (Chronos [2]), xLSTM (TiRex [5]), state space models (FlowState [13]), and MLP mixers (TTM [12]), all competitive on GIFT-Eval [1]. Parameter count is demonstrably not the lever there: Toto spans 4M to 1B parameters for 4.6 CRPS points, and sub-10M models beat far

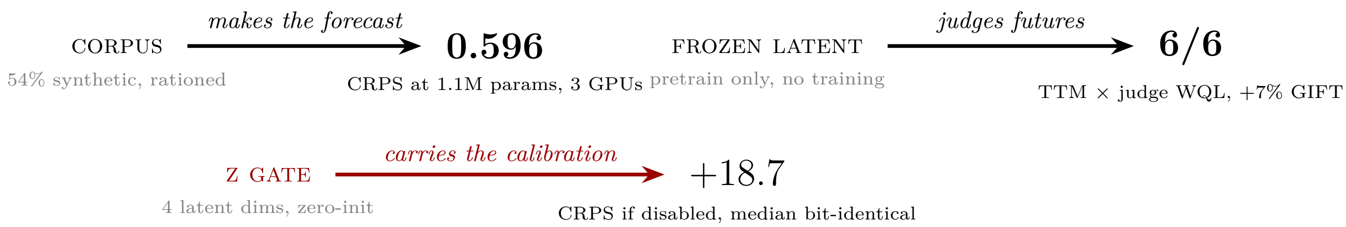

Figure 1: The three claims in three numbers. Data composition, not the pretraining objective, makes the forecast (Sections 5, 6); the frozen latent upgrades an external state-of-the-art point forecaster (Section 7); four predicted residual dimensions carry the entire calibration of the fan (Section 7, Figure 4).

larger ones. That is TimeJEPA's class, and why testing which pretraining objective serves forecasting at fixed small capacity has a place.

JEPA was proposed as a path to predictive world models [21] and instantiated for images and video [3, 4, 8], always with a disposable predictor; anti-collapse machinery comes from BYOL's EMA target [9, 14], VICReg [7], and LeJEPA's SIGReg [6], our default. Our energy reading of the latent follows LeCun et al. [22]. To our knowledge TimeJEPA is the first systematic attempt to keep the JEPA predictor as the forecasting organ for time series and measure whether that helps.

Four components carry load-bearing precedents: patching and channel independence from PatchTST [23], instance normalization from RevIN [19] with the robust arcsinh precedent in Toto [10], synthetic pretraining data from KernelSynth [2] via random Fourier features [24], and sign-flip TTA from TimesFM and YingLong [11, 27]. Remaining components and the methods entering our negative results are cited where used.

3 TimeJEPA

Figure 2 shows the model with its tensor shapes, Table 1 its measured parameter budget. Horizons beyond the native 256 use a rollout that propagates the full quantile fan under a comonotone coupling (Appendix B). Four design choices were forced by measurements; everything else, each detail with its own measurement, is in Appendix B.

- **Geometry randomization at fine-tuning time.** Skill initially peaked at the training context length and collapsed on both sides (+28.5% skill at context 384, −103.8% at 768 on one dataset): the model memorizes a patch count. Randomizing context and horizon per batch flattens the sweep from 33% spread to 5%; a scratch control attributes the robustness to fine-tuning-time randomization, not pretraining.

- **Robust normalization with a bounded inverse.** Inputs pass through $N(x) = \operatorname{arcsinh}((x - \operatorname{med}(c))/s(c))$ with a robust floored scale s, then RevIN [19]; the inverse clamps to a context envelope (precedents: Toto's scaler [10], Chronos's $\pm 15\sigma$ vocabulary [2]). What test-time symmetries survive this normalization is settled below.

- **A quantile head that attends back to the context.** The MSE-trained predictor converges toward a conditional mean (measured: predicted-latent variance 0.6 vs target 0.95), so dispersion cannot live in $\hat{z}$; the head cross-attends to context embeddings and decodes

$$\hat{q}_{\tau_j} = \hat{q}_{0.5} \pm \sum_{i \le j} \operatorname{softplus}(w_i), \tag{1}$$

  a fan that cannot cross by construction. Direct quantiles optimize the benchmark metric with no proxy; the median is the MASE-optimal point [17, 20].

- **A zero-initialized gate for residual statistics.** A 4-D head on the predictor trunk multiplies the widths of Eq. 1 by e^g, zero-initialized: exact identity at the start of fine-tuning, with consequences made precise below (Section 7).

Lemma 1 (affine quotient)**.** *Let $N(x) = \operatorname{arcsinh}((x - \operatorname{med}(c))/s(c))$ with s 1-homogeneous (ours, away from its numerical floor). For every $k > 0$ and b, $N(kx+b) = N(x)$: the normalization quotients the positive affine group. For the reflection, $N(-x) = -N(x)$.*

Component	Params	Share
Patch embedding	2,432	0.2%
Online encoder	595,072	52.4%
Predictor	419,968	37.0%
Quantile head	118,162	10.4%
RevIN affine	2	
Trainable total	**1,135,636**	
EMA target copy (frozen)	595,072	

Table 1: Measured parameter decomposition.

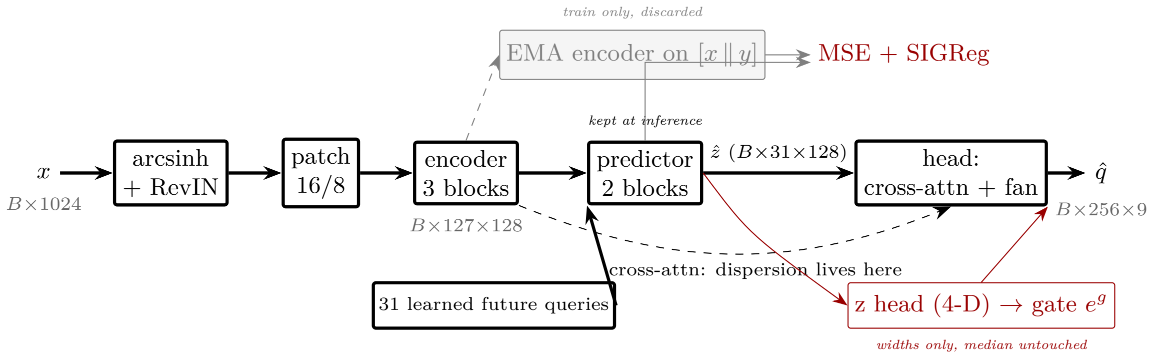

Figure 2: TimeJEPA with tensor shapes. Bold: the inference path, all of it kept, the predictor included (in I-JEPA it is scaffolding; here it is the forecasting organ), with the future represented by 31 learned queries, not teacher forcing. Gray: the pretraining branch, discarded at inference. Dispersion is produced by the head's cross-attention to the context, not by $\hat{z}$; the red gate multiplies fan widths only, leaving the median untouched (Proposition 1).

Proof. $\mathrm{med}(kx+b) = k\,\mathrm{med}(x)+b$ and $s(kx+b) = k\,s(x)$, so the argument of arcsinh is unchanged; under $x \mapsto -x$ it changes sign, and arcsinh is odd. □

Two consequences, both used: test-time scale augmentation is the identity (Table 2, pinned by a bit-identity test), and the sign is the only affine symmetry left, which yields the flip formula of Section 5.

Proposition 1 (median invariance)**.** *In Eq. 1 with gated widths* $\mathrm{softplus}(w_i)\,e^g$*, the median* $\hat{q}_{0.5}$ *does not depend on* g*. At fixed weights, any change in CRPS or coverage under gate surgery is therefore attributable to the spread, and MASE is structurally invariant.*

Proof. By inspection: g enters only the summed width terms. □

Pretraining is standard JEPA: MSE to EMA-encoded targets plus SIGReg [6] (a VICReg control was indistinguishable), AdamW at 3×10^{-4} cosine, bf16, batch 512 per GPU (Appendix B).

4 Corpus

Real data is LOTSA [29], split into univariate channels, with every subset matching an evaluation benchmark excluded by verified name patterns (all 28 GIFT-Eval datasets plus the Nixtla suite, whole datasets, pinned by a non-regression test); our patterns are strictly stricter than Salesforce's own sanctioned pretraining corpus at a quantified cost of 0.9% of observations (Appendix A).

Batches are drawn with square-root temperature weighting and a 3× oversampling cap, *rationed*: consumed greedily, the cap makes small families vanish early and batch composition drift through the epoch; converting it to a fractional per-batch quota makes composition stationary, and the rationed run produced the CRPS champion (Appendix A).

54% of the served batch is synthetic, a dose that is audited rather than estimated. It exists because the per-configuration error map showed holes exactly where no real data can be added: a frequency gap between 5-minute and hourly sampling, short yearly shapes rejected by chunking, and two morphologies with no public data at all, bursty IT-operations series and intermittent integer counts, generated by a random-Fourier-feature KernelSynth [2, 24] plus dedicated samplers. Shape augmentations follow the TiRex ablation [5]; scale augmentation is inert under the normalization (Lemma 1), so the budget goes entirely to shape.

5 Protocol and results

Protocol. One checkpoint, one configuration, all 97 configurations; nothing tuned or selected on GIFT-Eval data. The harness restates the official protocol next to its sources (measured convention drift: zero on MASE, ∼2.7% on CRPS); aggregates are geometric-mean ratios vs seasonal naive. Checkpoints are selected by intermediate evaluation, a doctrine forced by measurement: validation loss designated the wrong checkpoint three times out of three, once with a perfectly clean curve. The single test-time augmentation is the sign flip, $\hat{q}_\tau(x) = \frac{1}{2}[q_\tau(x) - q_{1-\tau}(-x)]$, the one symmetry Lemma 1 leaves; precedents include TimesFM, YingLong, and the 0.15M TinyCast [11, 27]. Table 2 shows why it is alone. We refuse multi-checkpoint ensembling, and report every aggregate plain and with flip.

Where the remaining gap lives. Two floors. The *tail*: sixteen configurations above ratio 1.25 cost about 12 aggregate points before the synthetic recipe compressed them to six, exactly as pre-registered; the residual tail is what the newest synthetic families target. The *body*: on the other 81 configurations we sit near 0.54 where the sub-10M leaders sit near 0.47, diffuse,

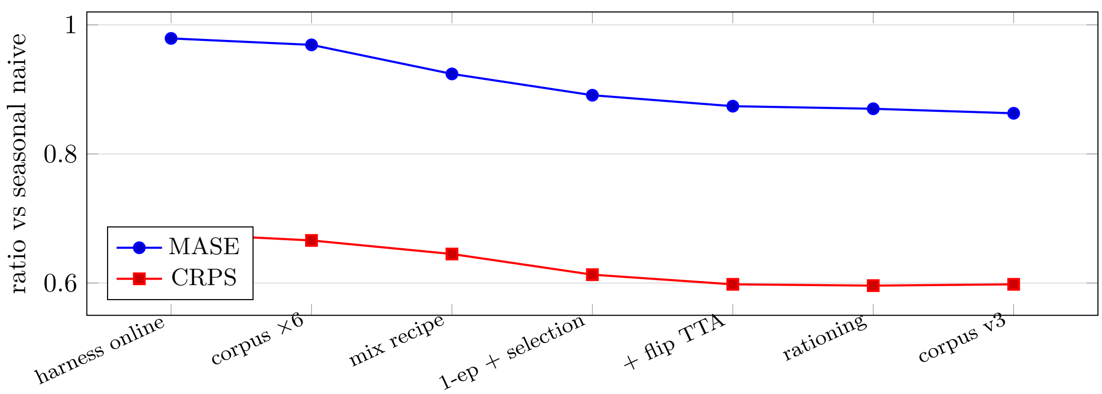

Figure 3: GIFT-Eval aggregates across the project, each step labeled by the single change responsible. Every step is a data or inference change; the objective contributes none, and volume alone is the weakest lever (corpus ×6 buys 1.6% CRPS). The last three CRPS points agree to ±0.002.

Variant	MASE	CRPS	Verdict
none (champion, plain)	0.895	0.624	reference
sign flip	**0.870**	**0.596**	official (2× cost)
scale $f(kx)/k$	=	=	identity (Lemma 1)
time shift (origins $-1..-7$)	0.898	0.627	staleness beats phase
multi-lookback	0.896	0.623	worse with flip
context 2048 at inference	1.190	0.876	must be trained

Table 2: Test-time variants: under a normalization that quotients the affine group, the sign is the only symmetry left.

no single culprit; term aggregates are flat, so the rollout is not the bottleneck.

A recipe ceiling, not a capacity ceiling. Three fine-tunes from three pretrains on three corpus recipes land at flip CRPS 0.597 ± 0.002: our levers are exhausted at this size. The cross-sectional evidence rules out raw capacity as the binding constraint (TinyCast reaches 0.545 with 0.15M parameters, FlowState-3M 0.496 [13]), and names the candidate walls: both sub-3M leaders train at long context (2048–4096) and treat time scale explicitly, and FlowState's own three-seed ablation prices those levers at +0.051 CRPS (sampling-rate equivariance) and +0.046 (parallel multi-horizon forecasts), together approximately our entire 0.094 gap. A ×3 model under the identical recipe, bands registered in advance, is in progress to measure the capacity share of the ceiling; the equivariance lever is the unfunded cross-resolution arm of Section 6.

6 The objective is not the lever

A reconstruction arm differing in exactly one variable (decode to raw normalized patches, SmoothL1) meets the latent arm at equal corpus, budget and geometry, checkpoints paired at 0.4% of validation loss: MASE 1.150 vs 1.166 on a seven-dataset benchmark, reconstruction winning 17 of 28 paired cells, and identical GIFT-Eval CRPS (0.979 / 0.6766 vs 1.003 / 0.6764). The objectives are indistinguishable.

Model	Params (M)	CRPS	MASE	Notes
TinyCast	0.1	0.545	0.774	
TTM-R3	1.4	0.520	0.724	per-regime configs
TimeJEPA	**1.1**	**0.596**	**0.863**	one config, flip
Lag-Llama	2.4	0.880	1.228	leakage flagged
Toto-2.0-4m	4.1	0.524	0.757	
YingLong-6m	7.3	0.609	0.880	
FlowState	9.1	0.502	0.726	
Kairos-10m	9.9	0.554	0.753	
Moirai2	11.4	0.516	0.728	
Moirai-large	311.0	0.599		adjacent rank
TimesFM-2.5	231.3	0.490		
Chronos-2	119.5	0.485		
Seasonal naive		1.000	1.000	

Table 3: The sub-12M class on GIFT-Eval (frozen snapshot; metadata is the leaderboard's own). TimeJEPA ranks ∼91st of 125 on CRPS, between FLAIR and Migas-1.0, ahead of Moirai-large, and seventh of the eight zero-shot sub-12M entries shown: mid-board overall, lower-third in its class, and the only entry trained on three consumer GPUs. The paper's claims do not rest on this rank.

The one signal favoring the latent arm, an advantage monotone in the GIFT term (+0.8/ + 3.1/ + 6.6% short/medium/long), dissolves under a per-step analysis: inside a window the gap *shrinks* with depth (slope −1.64% per 100 steps, 95% CI $[-2.82, -0.56]$, paired bootstrap over eight held-out datasets), while across horizons it tracks the number of autoregressive rolls (Spearman +0.404, $p = 0.019$). What survives is narrower: the latent objective may degrade less under iteration on its own outputs. A native h512 fine-tune, halving the roll count, gave the predicted sign at the magnitude of a minor lever (−1.7% on multi-rollout configurations; Appendix B).

One scope condition must be named: the duel is mea-

WQL vs SN ↓	ettm1	ettm2	etth1	etth2	weather	exch.
TTM-R3 alone	0.88	0.81	0.85	0.95	**0.63**	0.88
+ judge (centered)	**0.77**	**0.72**	**0.71**	**0.78**	0.64	**0.78**

Table 4: A frozen 1M judge turns a point forecaster into a probabilistic one that beats or ties its own WQL on 6/6 held-out datasets, zero additional training, replicated across judges from two pretraining lineages. On all 97 GIFT configurations (paired windows) the pipeline beats TTM's collapsed point CRPS by 7% at a 3.8% MASE tax; coverage (0.62 vs 0.80 nominal) is the open gap. Given a deliberately biased proposer, the judge moves its mass to the alternative center it keeps in the pool: a verifier that can fire its proposer.

Configuration (flip)	CRPS	80% cov.	MASE
champion, no z	**0.596**	0.769	0.870
z arm, best ckpt	0.598	0.790	0.874
z arm, early ckpt	0.626	**0.800**	0.921
gate bias → 0	0.626	0.813	0.921
z arm, *gate off*	0.785	0.968	0.873

Table 5: Residual-statistics ablations by evaluation-time weight surgery. Gate off: +18.7 CRPS points, coverage exploding, MASE moving by less than 10^{-3} (median invariance verified in production). Fine-tuning co-adapted the organs: base widths converge to a worst-case envelope, the gate to conditional sharpening; four latent dimensions carry the calibration. The counterfactual trained without z is the champion at 0.596: z buys nominal coverage at equal CRPS.

Hypothesis	Measurement	Verdict
Latent obj. beats reconstruction	null duel; within-window slope −1.64%/100, CI excl. 0	refuted as stated (§6)
Weight soup [30]	0.607 flip vs 0.596 single	average leaves the basin
Uniform conformal calib. [25]	$\gamma \approx 1$; CRPS +0.0002	fan already calibrated; zero-shot gap is *shift* (the z gate's job)
Scale TTA	bit-identical inputs	provable no-op
Translation TTA	0.627 vs 0.624	staleness wins
Long context at inference	0.876 CRPS	must be trained
E as planning regularizer	violations $78.7 \to 82.4\%$	classifier, not regularizer
Corpus volume alone	×6 data → −1.6%	weakest lever
Val. loss selects checkpoints	wrong 3/3	select by eval

Table 6: Negative results with their numbers (CRPS ratios, flip).

sured under this recipe's ceiling (Section 5), a regime where no objective may have room to differentiate; the claimed scope is this capacity, corpus and budget, and the ×3 rerun will widen it. Pretraining itself, independent of objective, buys transfer rather than fit: against a from-scratch control it is worth nothing in-domain (strict equality) and −8.7% MASE on datasets never seen at fine-tuning time. One architectural argument for the JEPA formulation survives untouched, unfunded: only latent comparison can impose resolution equivariance by the objective, since two resolutions do not share target values, and a competitor's ablation now prices exactly that lever at +0.051 CRPS [13].

7 The latent as judge

A pretrained JEPA is implicitly an energy over (context, future) pairs, $E(x, y) = 1 - \cos(\hat{z}(x), \mathrm{enc}([x \,\|\, y]))$ [22], lowered on true pairs by the invariance term. Everything below reads it off *frozen* checkpoints.

The judge exists; fine-tuning destroys it. Figure 5: pretrained checkpoints rank the true future at 0.21–0.25 among 34 candidates (chance 0.50; median rank 0.000 on hourly electricity for the best judge), while a fully fine-tuned checkpoint degrades to 0.41. The judge peaks early in pretraining and erodes while validation keeps improving, so judge selection uses an early probe; it also scales with capacity faster than forecasting does.

A zero-training reranker that upgrades TTM-R3 (Table 4). Candidate futures are scored with E and read out as weighted quantiles; candidates share the proposer's center, and degenerate pools fall back to uniform weights (pool design, and the four-fold dilution replication that forced it, in Appendix B).

Four dimensions carry the calibration (Table 5). The predictor converges to a conditional mean (Section 3); rather than the residual, irrecoverable by definition, a 4-D head predicts its *statistics* (log-RMS, log-MAD, skewness, lag-1 autocorrelation around a causal EWMA), deterministic functions of the data, so nothing can collapse. Predicted volatility correlates with realized at 0.63, against 0.3–0.4 for the signal path: volatility is more predictable than its realization. Figure 4 shows the mechanism on one window.

8 Clean negative results

Each row of Table 6 was a pre-registered hypothesis worth building; several contradict folk practice. Expanded discussion in Appendix B.

9 Limitations and outlook

One seed per arm everywhere, mitigated by paired designs and pre-registered directional predictions, not solved. The 0.597 ± 0.002 convergence is a ceiling of this recipe, not of the parameter count (Section 5); the

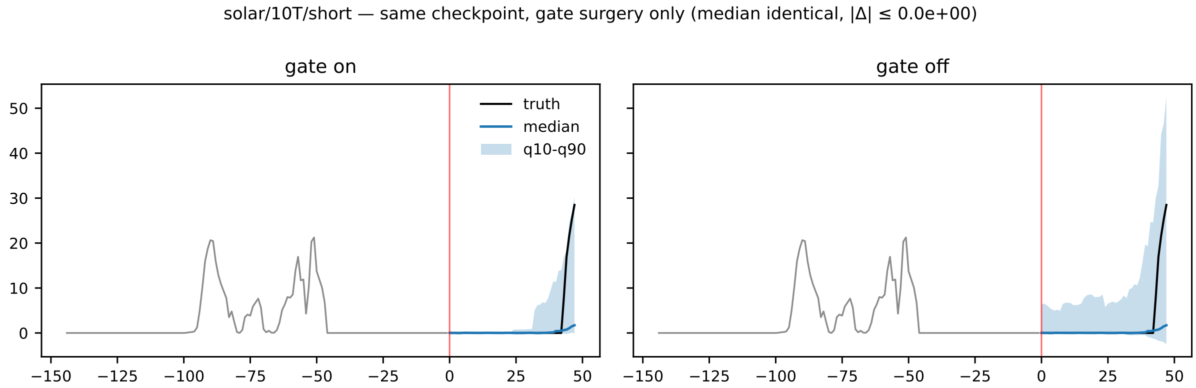

Figure 4: The gate on one solar window (10-minute data), same checkpoint, same median (max difference 0), gate surgery only. With the gate, the fan is pinned to zero through the night, where the outcome is certain, and opens at sunrise; without it, the same weights spread a worst-case envelope over certain zeros. Note that the truth briefly escaping the band at the peak is what a calibrated 80% interval is supposed to do one step in five; the gate-off panel that covers everything is the 0.968 coverage pathology of Table 5 in picture form.

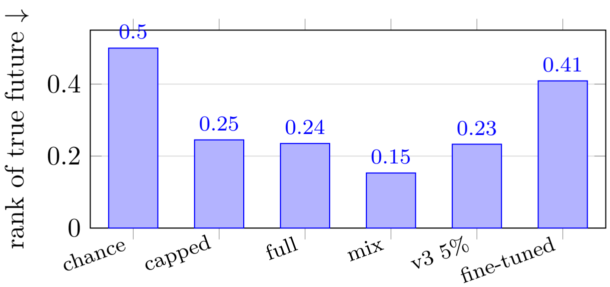

Figure 5: The judge lives in the pretrain and dies in the fine-tune (normalized rank of the truth among 34 candidates; 0.50 is chance).

highest-priced named levers, trained long context and explicit time-scale handling, are future work, the latter being the unfunded cross-resolution arm. Ranks are against one frozen leaderboard snapshot, through a harness with measured convention drift. The hybrid's coverage gap (0.62 vs 0.80 nominal) is open. The model is univariate by choice.

The judge results point past forecasting: a zero-initialized action input (the FiLM pattern of Section 3, identity at $a = \varnothing$) would unify forecasting and predictive control in one arm, with planning prudence supplied by the fan and the z head rather than by the energy, which an owned-simulator study showed to be a classifier blind to state continuity, not a regularizer. The wall is causal data, not architecture; the design is in Appendix D.

10 Reproducibility

Every number traces to a dated experimental log whose entries register hypotheses and numeric success and failure bands before each run; configurations are declarative one-variable overlays; evaluation results are cached per configuration and keyed by a checkpoint fingerprint; the corpus assembles by symlink from immutable, seeded sources with an audited batch composition. Total compute, every failed run included: a few GPU-weeks on three RTX 3090s. The 295-test suite pins mechanisms that once failed silently (Appendix C).

Acknowledgments

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Family	Morphology	Target hole	batch share
ops_bursty	Poisson bursts, exact zeros, saturation	10-second IT-ops (no public data)	19.7%
intermittent	integer counts, neg. binomial sizes	sparse monthly demand	9.9%
subhourly	periods 24–150	10/15-minute frequency gap	7.4%
broadband	periods 4–2048	diversity floor	7.7%
lowfreq	periods 4–52, high trend prob.	short-series shapes	5.7%

Table 7: The five synthetic families (audited v3 shares; 54.0% synthetic total).

A Corpus and batch composition in detail

Chunking and windowing. Series are segmented into fixed-length, non-overlapping float32 chunks (object arrays break copy-on-write under forked data workers; measured as a 51 GiB resident set), chunk length adapted per subset and capped at 2048 for real data. Chunks above 5% missing are dropped; below, gaps are linearly interpolated (rejecting any NaN cost 100% of two metro datasets whose nightly closures are regular; a higher threshold would fabricate 3 a.m. ridership). Training windows slide at stride 8 over a flat int32 index. A per-family cap prevents climate reanalysis (63 of 118 audited files) from becoming half the corpus. Short-series configurations get a pad-to path: short real series are left-padded by repeating their first value, so the real data occupies the end of the chunk where the target is read; the target is always real, and the flat-then-data context is the condition evaluation already imposes. Geometry still binds: a 256-step real target cannot come from a 30-point yearly series. Real 5-minute data is additionally mean-pooled by 2 and 3 into legitimate 10- and 15-minute series (downward aggregation is physically correct; upward interpolation would fabricate information).

Sampler drift and rationing. Family i receives batch share $\propto n_i^{0.5}$ per *file*, with an oversampling cap of 3× its natural share per epoch (at 6×, small files overfit visibly). Per-file weighting means a family sharded into k files gets k draws; we first suffered this (climate shards inflated to 66.5% of the batch until capped) and later exploited it (the synthetic dose is set by shard count: 3 unsharded families measured 11.2% of the batch, the 23-shard v3 allocation was predicted at ∼51% and audited at 54.0%). Consumed greedily, a family's cap exhausts early and the family vanishes: audited, 16 of 71 families extinct before 1% of the epoch, capped-family share drifting from 46.6% to 35.7% between first and last decile, batch shrinking from 493 to 409 as slots went unfilled. Rationing converts each cap into a fractional per-batch quota with carry-over (a family entitled to 0.4 samples per batch contributes 2 every 5 batches): same total exposure, composition stationary across all ten deciles. Augmentation cannot substitute for it: it diversifies the samples that are present, not the families that are absent.

The synthetic generator. Exact GP sampling is $O(N^3)$; summing K=64 sinusoids drawn from the kernel's spectral density [24] is $O(NK)$ and indistinguishable by eye at 8192 steps. Component bank: seasonal blocks with decaying harmonics, smooth GPs (length scale log-uniform in $[16, 1024]$), trends with 30% slope breaks, Student-t noise 15% of the time, level shifts and impulses, 25% multiplicative modulation, and a final random affine map so synthetic series are not recognizable by their normalization. Table 7 lists the five families. The ops_bursty family draws Poisson-arriving bursts with amplitudes up to 300× a log-uniform base level and exponential decay, zero-inflation segments at exactly zero, capacity saturation at a high quantile, and grid quantization; the intermittent family draws integer counts from a Bernoulli occurrence process with negative binomial sizes. External calibration: Chronos reports about 10% synthetic as optimal in their setting [2]; Toto-2 reports a majority synthetic recipe [18]; our 50–55% batch target sits deliberately at the aggressive end because the per-configuration error map showed the holes to be exactly where no real data can be added. The dose is audited, never estimated: a first estimate landed at double the intended share and was corrected before the run.

Augmentations. Shape-changing transforms only (scale augmentation is provably inert under the normalization): amplitude modulation by a random piecewise curve (p=0.5); censoring at a random high quantile with the same threshold for context and target, since saturation is a property of the world (p=0.5); periodic spike trains applied to both (p=0.05; a random spike in the target alone is unlearnable noise, a periodic one is signal). Order mirrors physics: modulate, spike, then censor. Mild jitter is context-only, keeping the JEPA target clean.

Accounting. The final corpus assembles by symlink from immutable sources: capped LOTSA (10.05B observations, 4.9M chunks), extra public datasets, the padded short-series block, the decimated block, and 23 seeded synthetic shards (cross-corpus seed collisions impossible by derivation). The audit is the corpus's identity card: 106 families, 54.0% synthetic, capped real tail 24.6%, composition stationary across deciles. One imprecision reported rather than hidden: the pre-v3 capped corpus was logged as both 3–4 billion and about 1.7 billion observations; the exact count was never instrumented, and the volume-scaling comparison in Section 5 states the range.

B Architecture, training and evaluation details

Geometry. The geometry bundle (context 1024, native horizon 256, patch 16/8, randomization at fine-tuning) came from a dedicated round that improved a seven-dataset average MASE from 1.365 to 1.159 and flattened the context sweep (512 to 1280 within 5%, against 33% before). Randomization is per batch, keeping tensors rectangular with no masking; validation always uses native geometry so losses stay comparable.

Encoder, target, predictor. Three pre-norm blocks [32] with RoPE [26], 4 heads, feed-forward 512; the target encoder is an EMA copy on the BYOL/MoCo v3 cosine schedule, $\tau : 0.996 \to 1$ [9, 14]. The predictor concatenates 127 context embeddings with 31 learned future queries; the query table is deliberately oversized because slicing past a table's size silently returns fewer rows, a failure that once corrupted every horizon above 136 steps with valid shapes throughout (Appendix C). Horizon extension at fine-tuning grows the table by merging trained rows. An optional zero-initialized FiLM conditions the queries on a scalar, built for cross-resolution training and reusable for action conditioning.

Anti-collapse and two lessons. SIGReg [6] at λ=1 (16 random 1-D projections toward Gaussian); a VICReg arm was indistinguishable (1.193 vs 1.183 average MASE). Two implementation lessons cost real results: variance statistics must be computed per patch position, not pooled (pooled variance 0.9990 against per-position 0.0000, positional diversity masking true collapse at fixed positions), and a regularizer computed on a detached tensor contributes zero gradient while inflating the reported loss.

Optimization budgets are measurements. The 3×10^{-4} ceiling: at higher rates the online encoder outruns the EMA teacher and latent targets drift. A cosine calibrated for 40 epochs never decays in 3 (the run oscillated and its best checkpoint was a noise trough); the full corpus is exhausted in about one epoch, after which validation rises. Collapse monitoring (per-position std, effective rank) must be calibrated per corpus: rank at initialization is 3.7–4.8 on a 24-dataset corpus and 43–87 on LOTSA.

Comonotone fan rollout. Feeding back the median beyond the native horizon has a measured bias: the median path is smoother than any trajectory, so intervals shrink with depth (a 10–90 band of width 0.267 at h720 against a truth growing like $\sqrt{h}$). We propagate all nine levels, copy k continuing from its own level k; the marginal fan is the per-step sort. The assumption, stated honestly, is perfect rank dependence across rolls. Deterministic, no sampling, and it moved the average fan gain at h720 from -4% to -16% WQL with band widths finally increasing in depth.

The h512 arm. A native h512 fine-tune (grown query table, same pretrain) plateaus at 0.892 / 0.612 with flip. Its deficit decomposes mechanically: the 1536-step training window drops the decimated 10-minute block and the padded short-series block from the fine-tuning corpus, costing +11.1% on the 21 affected configurations, while unaffected multi-rollout configurations improve by 1.7% relative to single-roll ones (Section 6). Its 40% checkpoint is the best-calibrated of the project, empirical 80% coverage of exactly 0.800 (0.102/0.901 at the nominal 0.1/0.9), plausibly because horizon-randomized fine-tuning up to 512 steps teaches the head to calibrate across horizons.

Reranker pool design. The propose-judge-reweight pipeline pools the proposer's clean path and jittered variants, seasonal naive as an alternative center, and K bootstrap candidates, softmaxing standardized energies into weights. Two measurements fixed the design. Bootstrapping *raw history* centers candidates on the past, and every gram of weight they receive pulls the weighted median off the proposer's trend whenever the proposer is right: this dilution signature replicated four times, on the same two datasets, across three different proposers and two judges. The fix bootstraps seasonal innovations $y_t - y_{t-m}$ and re-attaches the blocks to the proposer's path, so every candidate shares the proposer's center and dilution of the center is impossible by construction; registered predictions (both dilution datasets recover, all four wins keep their margins) held. Separately, a proposer-paths-only pool degenerates because a point forecaster contracts input jitter: near-identical candidates give near-identical energies, and dividing by a tiny energy standard deviation turns the softmax into a noise amplifier that dumps mass on anchor candidates (measured as localized disasters on daily and weekly configurations where a linear drift anchor diverges, and an 80% coverage of 0.206). Energies below a variance floor therefore fall back to uniform weights, and the drift anchor left the pool. Candidate refinement by gradient descent on E repairs the envelope of uncentered pools (replicated) but is null to slightly harmful on centered ones: refinement and centering are substitutes, and the pipeline uses centering alone. Judging power also scales with capacity faster than forecasting: a $\times 3$ model at 5% of one pretraining epoch already out-judges the fully pretrained small model on both the probe (0.211 against 0.233) and the GIFT reranking task.

Negative results, expanded. The weight soup's damage is localized (one weekly configuration's ratio moving from 2.76 to 4.13), the signature of interpolating between unaligned solutions. The conformal no-op's gammas, $\gamma \approx [1.01, \ldots, 1.12, \ldots, 1.03]$ calibrated strictly on held-out training data, show the fan already conformally calibrated in distribution, which relocates the observed zero-shot under-coverage to distribution shift, precisely the conditional problem the z gate addresses. Long context taken at inference alone collapses hourly configurations (unseen patch counts) even though two high-frequency blocks improve dramatically; long context must be bought at training time. Test-time translation loses more to staleness than it gains in phase decorrelation.

C Silent failures, and how they were actually caught

Most defects that invalidated results were silent: valid shapes, decreasing losses, no warnings. Three of the worst were caught by reading log lines, not by tests. Table 8 is the catalogue; we consider it as reusable as any architecture choice.

Defect	Consequence	Caught by
Eval-side global normalization vs per-instance training	MSE overstated 42% avg over 40 pairs; nothing pre-repair usable	protocol audit
RevIN affine inverted at decode, never applied forward	6–10% scale error + offset on every forecast	code reading
Query table sliced past its size	missing queries silently replaced by context embeddings, trained as predictions; every horizon > 136 corrupted	code reading; now refused
VICReg pooled over positions; half on a detached tensor	positional diversity masked true collapse; zero-gradient regularizer	measurement (0.9990 vs 0.0000)
Optimizer built before gradual unfreeze	unfrozen weights never stepped; every gradual run was a frozen-encoder probe	log line: parameter counts
Augmentations parsed, never wired	first-phase runs trained without their declared augmentations	log line
`last.ckpt` = last top-k copy, not last state (Lightning version difference)	mid-run checkpoint served as "final"; the operator had measured it, review dismissed it from another version's source	18 bit-identical probe values; md5. Measurement beats reading code. Parade: save every checkpoint
Corpus contained the benchmark	electricity/traffic-hourly extend the Nixtla series over their eval period; early transfer claim downgraded	found while writing exclusion rules for another corpus
Cached eval resumed across an overwritten checkpoint	old model's numbers reported as new, "already done" $\times 97$	fingerprint now in cache key

Table 8: Ten defects, none caught by the test suite that now pins them.

The pattern: correctness of mechanism and coherence of protocol are different properties. Tests grew to pin each mechanism after the fact; what remained was protocol-level and was caught by two near-free habits. Pre-registered numeric expectations: a result that contradicts its prediction triggers a search that a merely good-looking result never would. And reading the run's own log lines (loaded-key counts, skipped-file lists, per-family batch shares) instead of trusting dashboards.

D Design for an actioned world model

The judge results reframe the predictor as a world model of an autonomous system; this appendix records the design its measurements fix, as future work.

Mechanism. An action embedding conditions the predictor through zero-initialized FiLM, the pattern of Section 3: $a = \varnothing$ is the identity, so the actioned model *is* the current forecaster with unchanged benchmarks. A spectator has no action, which is marginal dynamics, already learned; an actor knows its candidate action by definition. The controller is not a network but inference-time optimization, planning by backprop with the action as the variable, or propose-and-judge over candidate plans [15, 28]. The predictor never decides; it simulates, and the loop decides.

What the judge knows: the thermostat study. On a fully owned testbed (a linear thermostat with hidden action, history generated under a bang-bang policy, so that every verdict is checkable against the true dynamics), the frozen judge discriminates time-reversed futures (AUC 0.808) and morphology violations (0.935) but is blind to state-jump discontinuities (0.556, chance): it knows the arrow of time and the shape of trajectories, and accepts a teleported future. Consistently, using E as a planning regularizer fails a fair test: the dormant term changes nothing and the active term raises violation rates from 78.7% to 82.4% across three concordant runs. The energy is a classifier, not a regularizer.

Constraints this fixes. Prudence comes from the fan and the z head, not from the energy, converging with the uncertainty-regularized MPC of Henaff et al. [16]. State continuity must be an explicit term of the planning cost, since the energy will not supply it. The judging checkpoint is not the forecasting checkpoint and is selected by an early probe.

The wall is data. Observational corpora carry no labeled actions and are confounded: a model trained on thermostat logs learns that heating turns on when temperature drops, and a naive planner cuts the heating to warm the room, a failure we measured in miniature on the owned simulator (plans pulled toward historical setpoints). The exit is what the synthetic pipeline makes cheap: causal generators with labeled interventions [31] grafted onto the corpus machinery. The through-line: an unobserved action is a latent variable explaining the future, the z of LeCun [21]; the residual-statistics head is its deterministic shadow, the action input its declared form.

The claim this aims at. A quantile-fan forecaster whose native output supports chance-constrained model-predictive control, zero-shot, on unseen systems, with $a = \varnothing$ leaving its benchmarks untouched. Every component (fan, gate, judge, refinement machinery, generator pipeline) exists in this codebase; the sequence is action FiLM, causal corpus, then the planner that is already written.